DASC 3240, L09

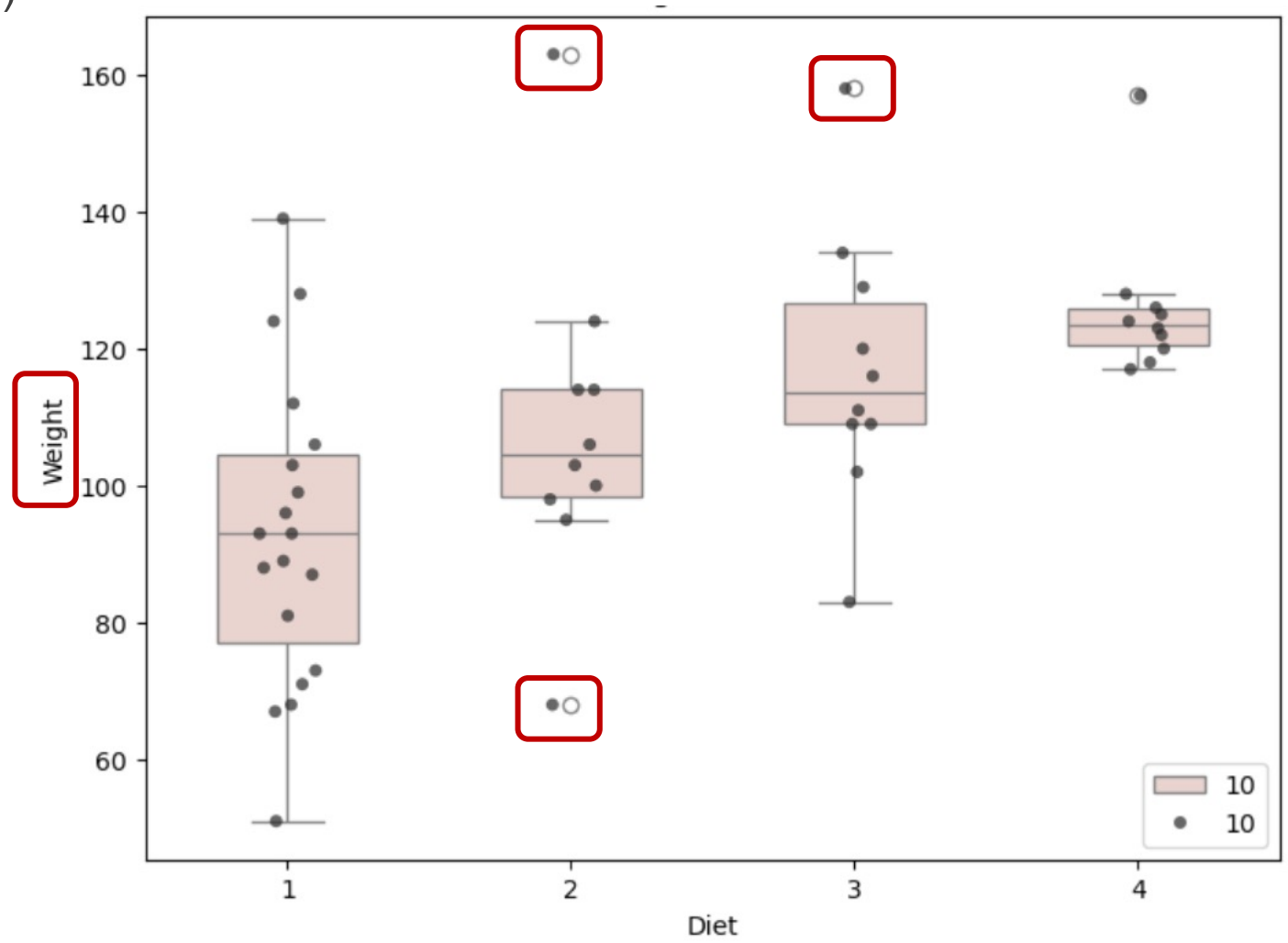
Plotting two variables – Scatter plot

2023/24 SPRING SEMESTER

MASAYUKI USHIO, ASSISTANT PROFESSOR AT OCES
USHIO@UST.HK (RM CYT-2013)

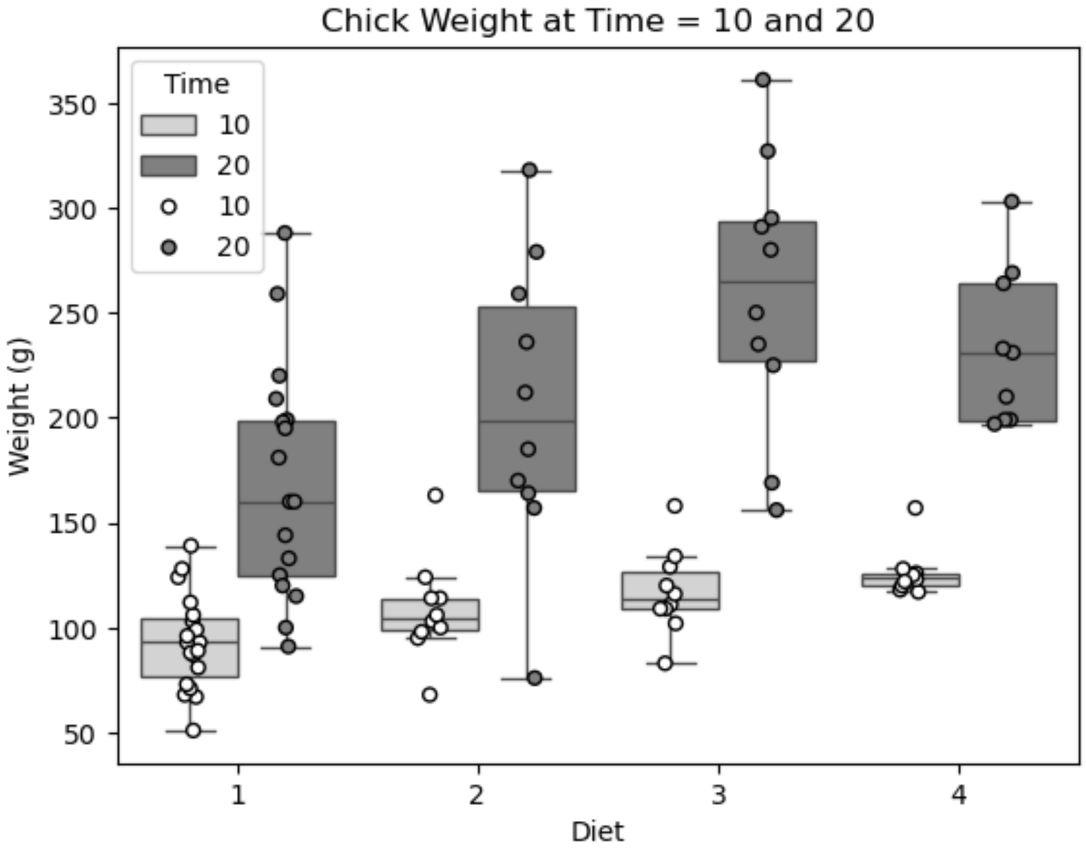
Feedback about the assignments last week

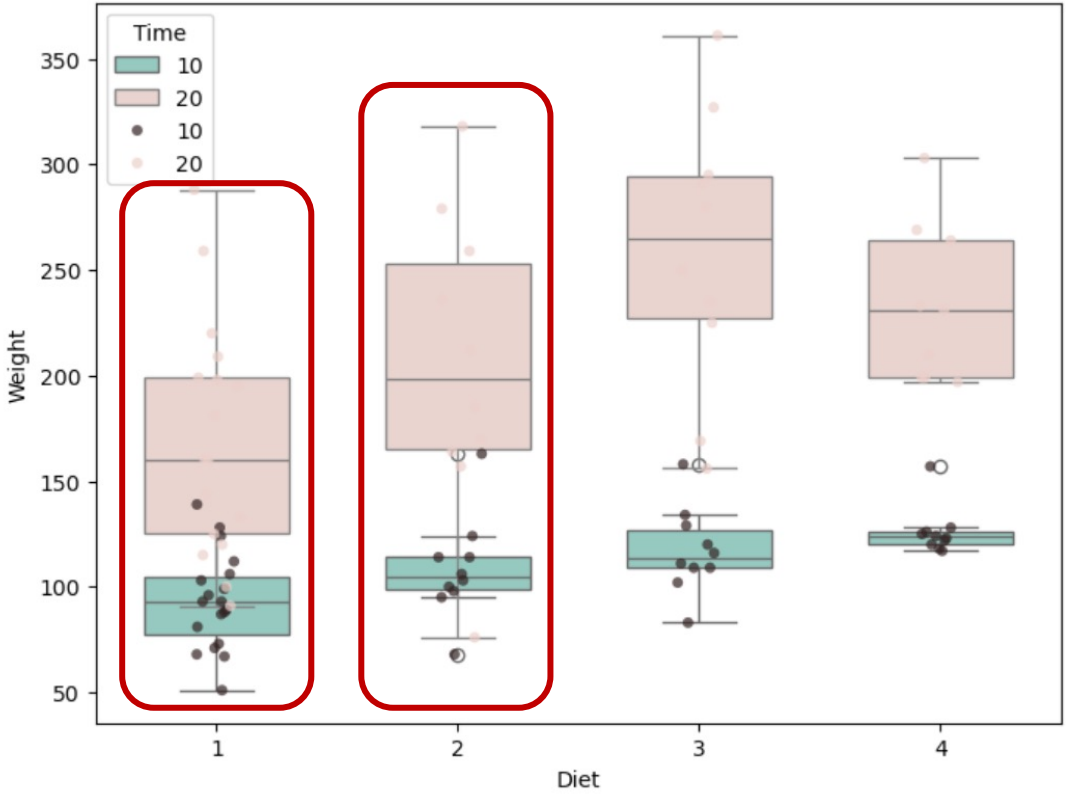
- Add a unit to the axis label(s)
- Avoid duplicate outliers



Feedback about the assignments last week

- Do not use the same slot when you want to show different categories. This is not good especially when data points overlap.





Today's outline

- Plotting two variables
  1. Scatter plot
  2. Regression line
     - Linear regression
     - Nonlinear regression (LOWESS)

Plotting two variables

TWO NUMERIC VARIABLES

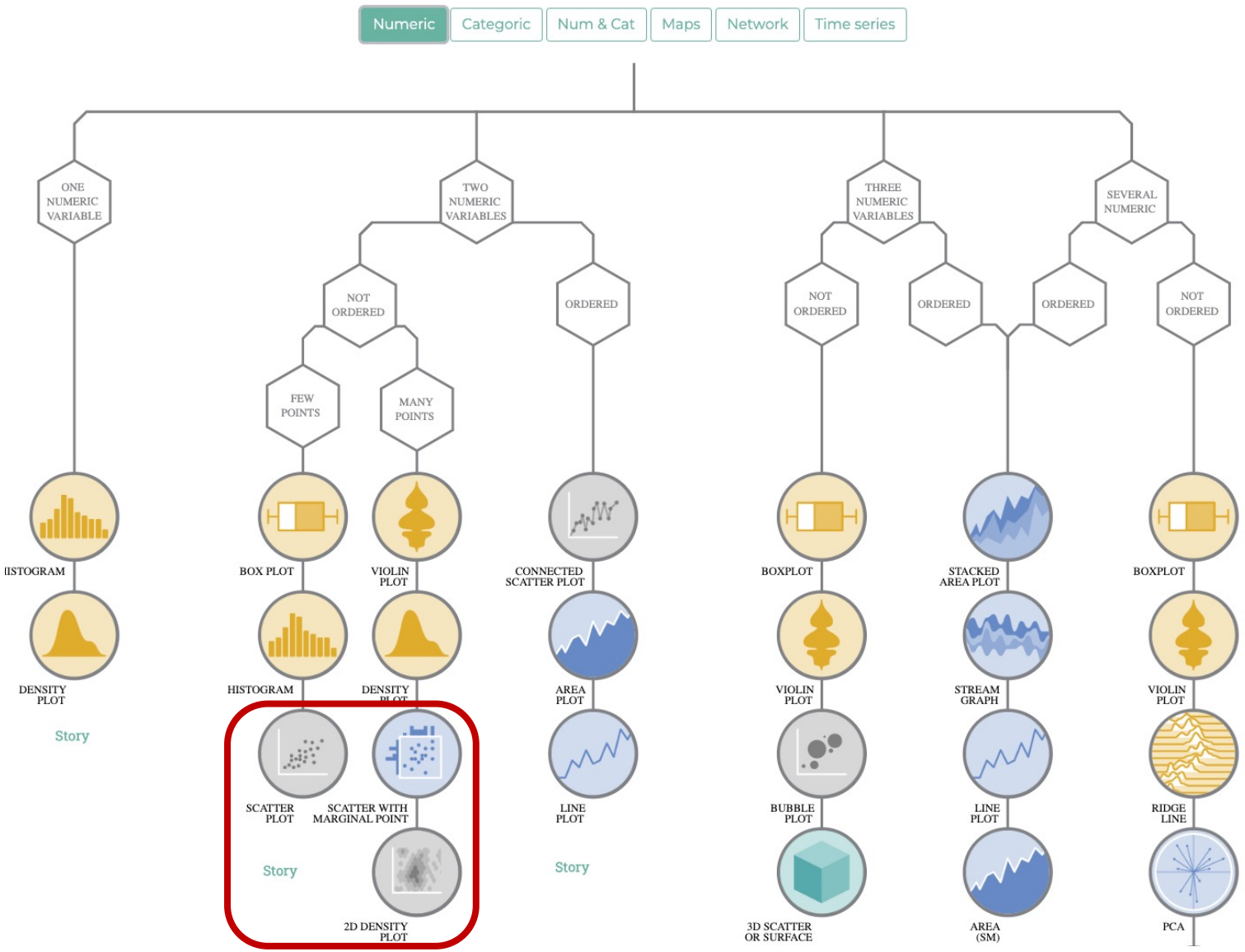
What should we consider when we visualize data?

1. The number of variable
2. Is the variable(s) numeric, or categorical?
   - **Two variables – Numeric & Numeric**
     1. (Barplot, boxplot, jitter plot, ...)
     2. Scatter plot

How many variables? Numeric or categorical?

Numeric or categorical?

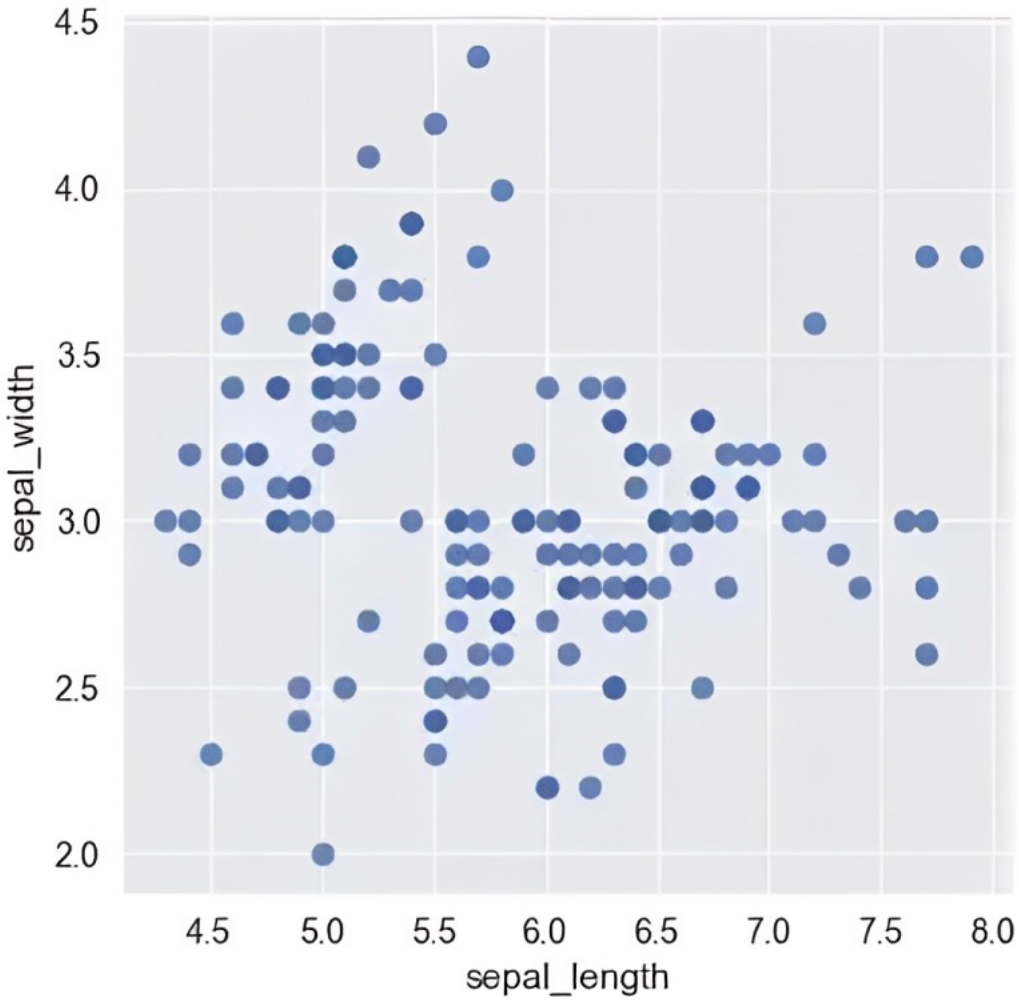
The number of variables



https://www.data-to-viz.com/

Scatter plot (scatterplot)

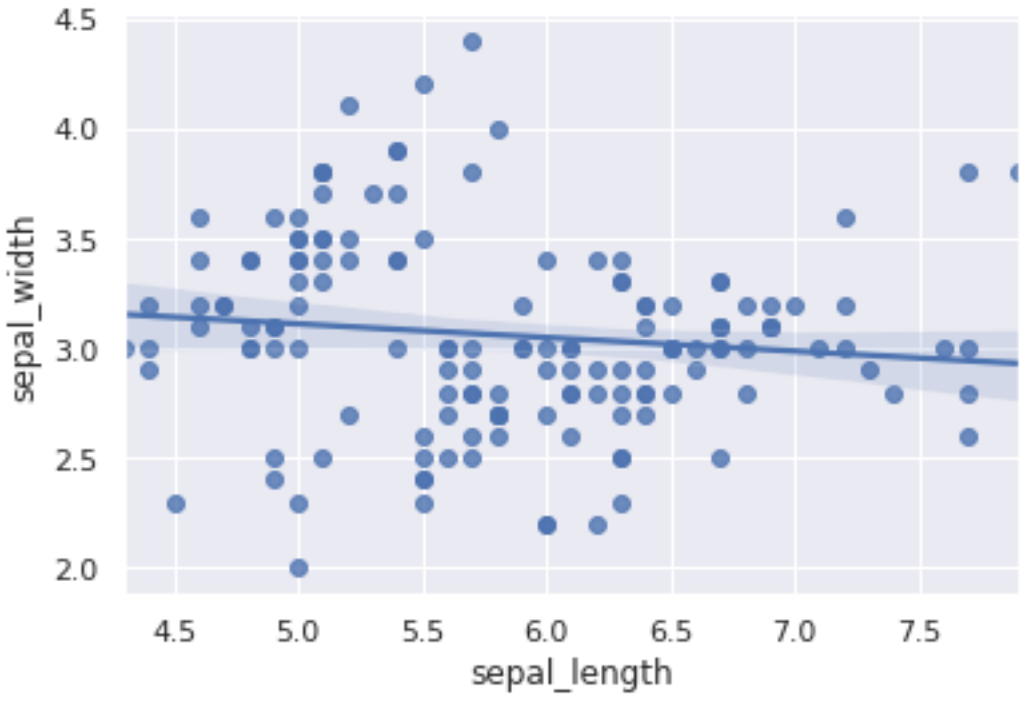
- A **Scatter plot** displays **the relationship between two numeric variables**, one being displayed on the *X*-axis (horizontal) and the other on the *Y*-axis (vertical).
- Each data point is often represented as a circle, but depends on the number of categories shown.
- One of the most common way to visualize the relationship between two variables.



https://python-graph-gallery.com/scatter-plot/

Scatter plot with a regression line

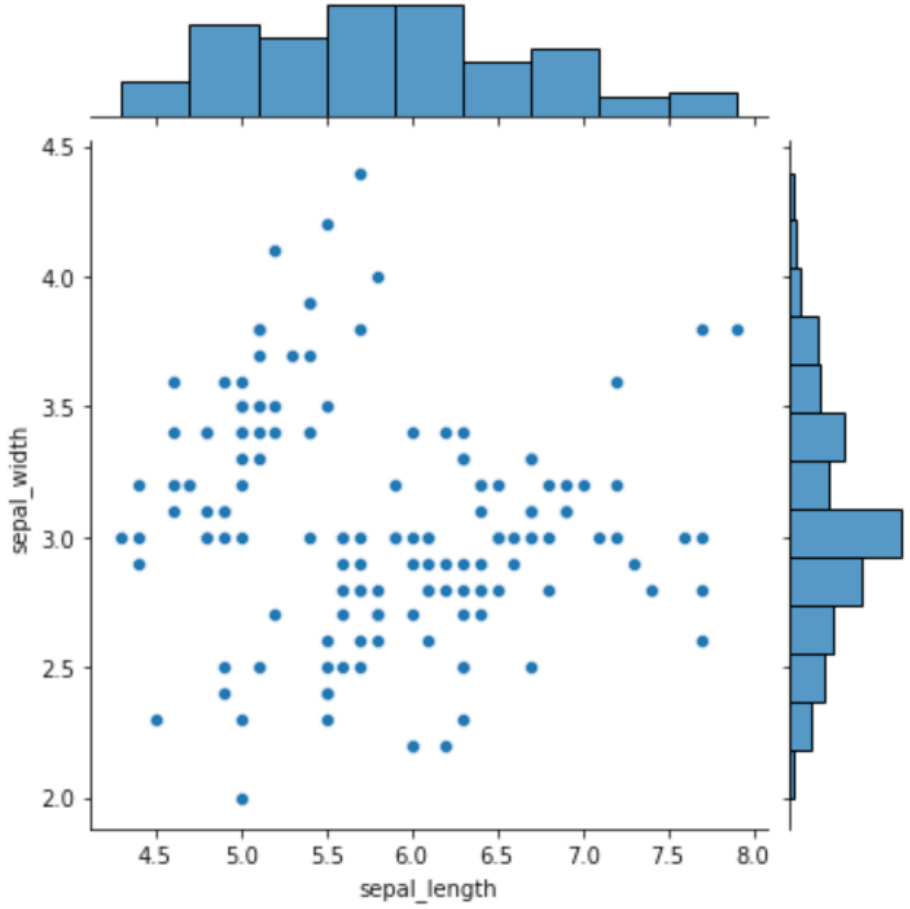
- A **regression line summarizes** the relationship between two numeric variables (**a blue solid line** in the right figure), which helps easy interpretation of the relationship.
- Drawing a regression line requires **statistical analysis**, and there are many methods to draw a regression line.
- A regression line is often drawn with **a confidence intervals** (**transparent blue area** in the right figure).



https://python-graph-gallery.com/scatter-plot/

Scatter plot with other graphs

- A scatter plot can be customized by combining it with other plots.
- For example, two histograms are plotted along the the X- and Y-axis in the right figure, visualizing the distribution of each variable.

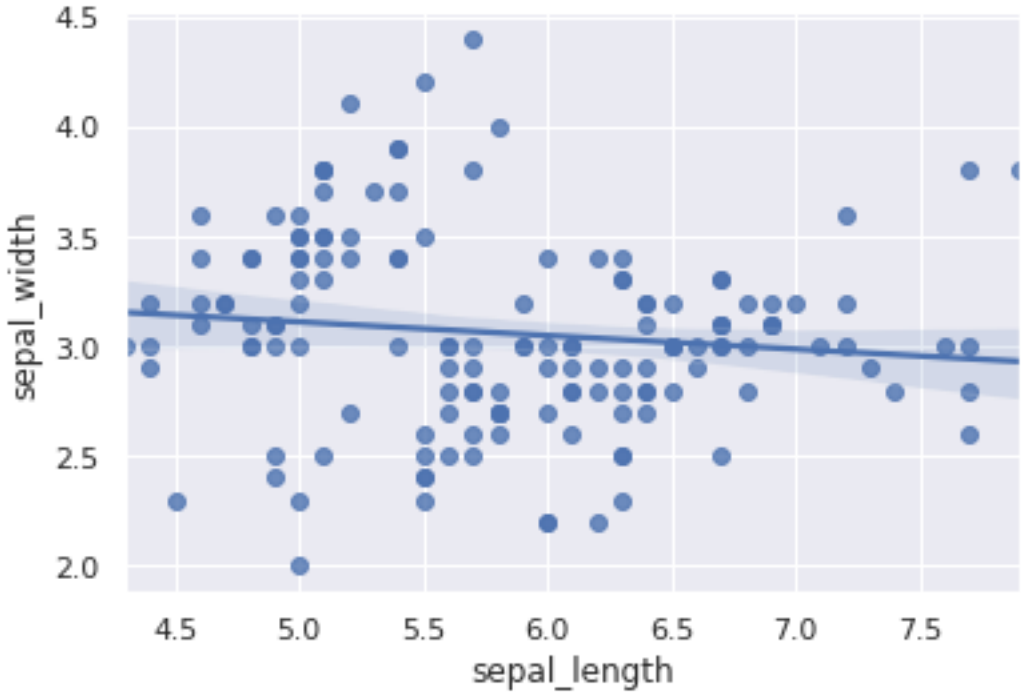


Statistical background of regression

LINEAR AND NON-LINEAR REGRESSION

Regression

- Drawing a regression line and confidence intervals using a package is **very easy**, but **understanding its algorithm requires statistical knowledge**.
- Explaining detailed statistical backgrounds of various regression methods is beyond the scope of this course, but I will introduce some.

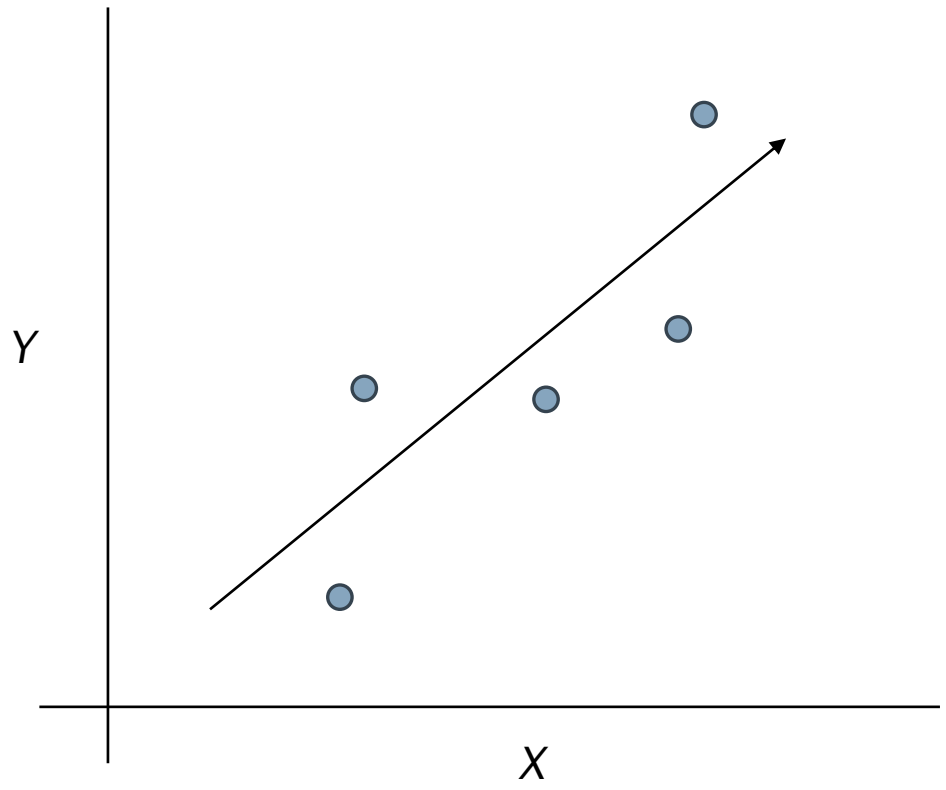


Interpretation of Cis depends on the algorithm to generate them, but for example, we could say that 95 times out of 100 bootstrapped trials (e.g., randomly select 50% of the original data, and repeat it 100 times) generate regression lines fall within the shaded area.

https://python-graph-gallery.com/scatter-plot/

Linear regression

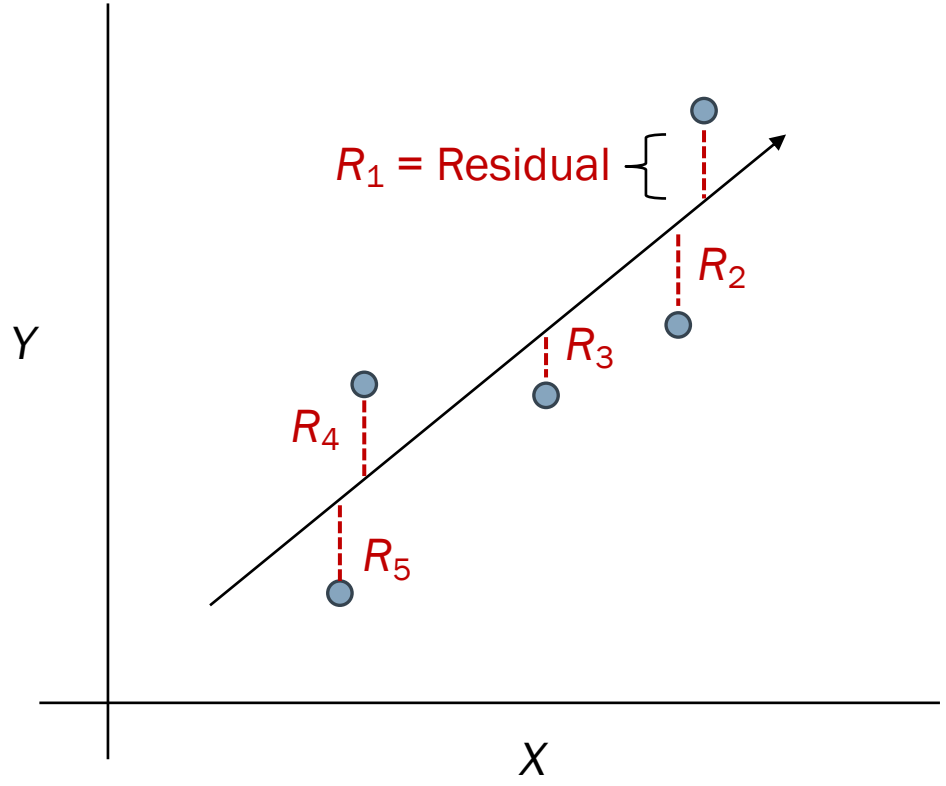
1. **Assume** there is a linear relationship between two numeric variables.
2. Typically, $Y = aX + b$
3. Estimate the most appropriate coefficients, a and b, using **a parameter estimation method**.



Linear regression

- The simplest method is "**least squared**" method.
- Calculate the distance between a point and the regression line. If we have five data points, we can calculate five "**residuals**".
- We can "**square**" each residual and determine a and b so that the sum of least squared values is minimum.

$$SS = R_1^2 + R_2^2 + R_3^2 + R_4^2 + R_5^2$$



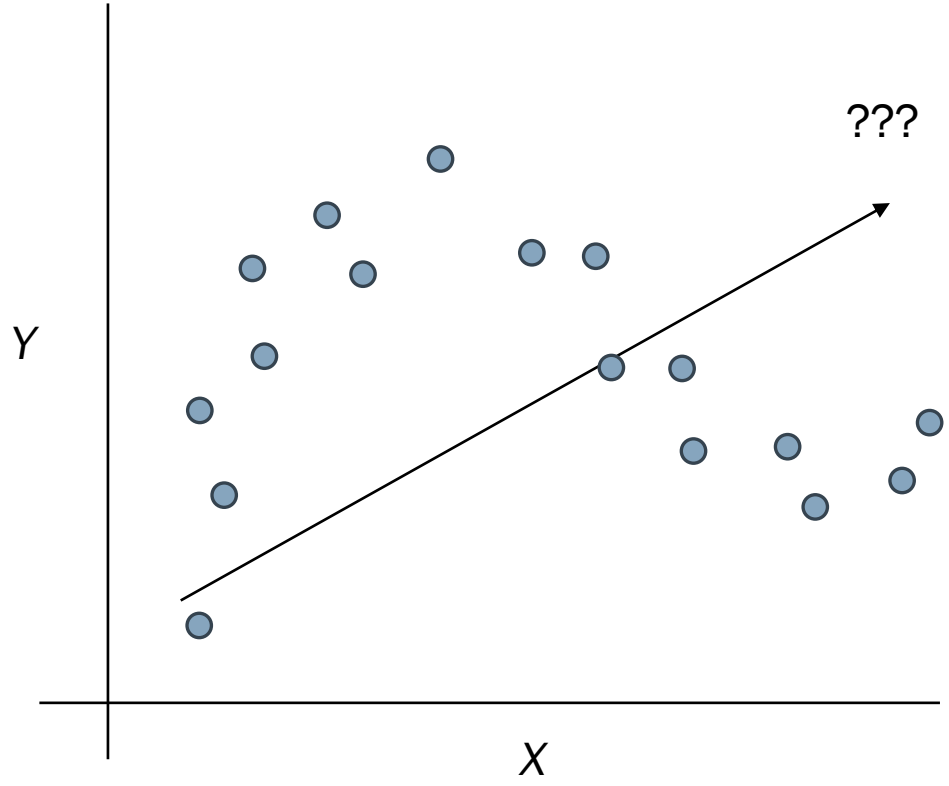
Linear regression

StatQuest (https://statquest.org/)

https://youtu.be/nk2CQITm_eo?si=-fPjIJ6KZOdW6SZ1

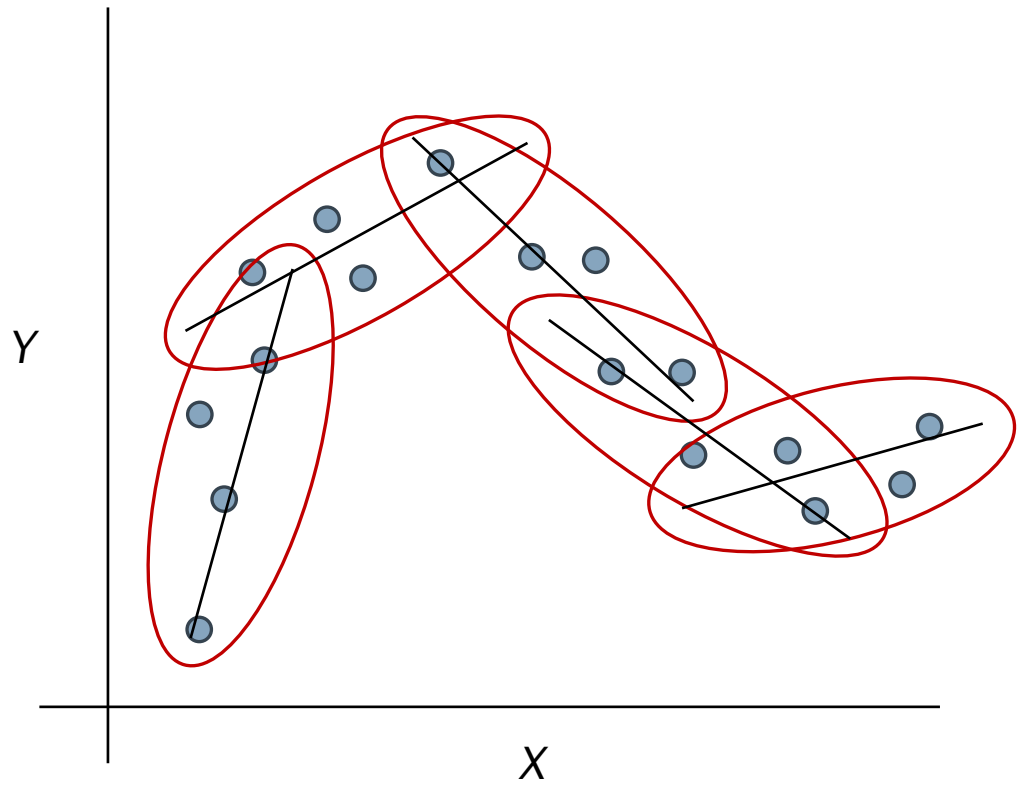
Fitting a curve

What should we do if we **cannot** assume a linear relationship?



Fitting a curve: LOWESS fitting

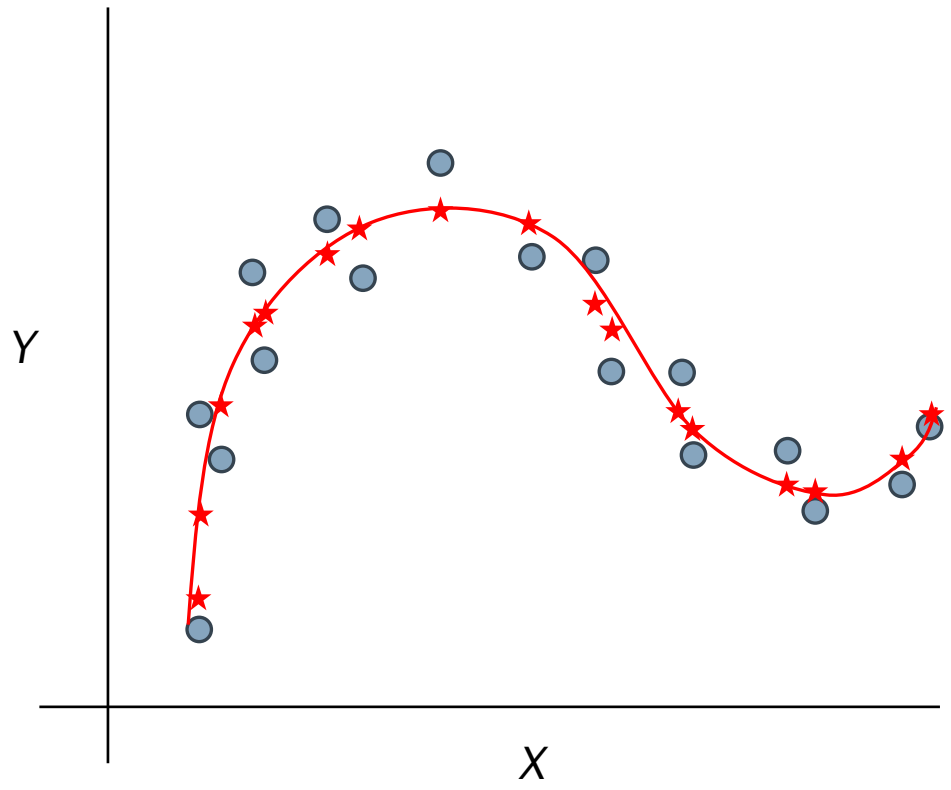
1. Use a type of **sliding window** to divide the data into small subsets.
2. Perform a (weighted) linear regression in each window. Alternatively, you can try a (weighted) polynomial regression.



Fitting a curve: LOWESS fitting

1. Use a type of **sliding window** to divide the data into small subsets.
2. Perform a (weighted) linear regression in each window. Alternatively, you can try a (weighted) polynomial regression.
3. Each predicted value (★) is calculated for each data point (= each window), which will result in a smoothing line.

Locally Estimated Scatterplot Smoothing (**LOESS**) or Locally Weighted Scatterplot Smoothing (**LOWESS**)



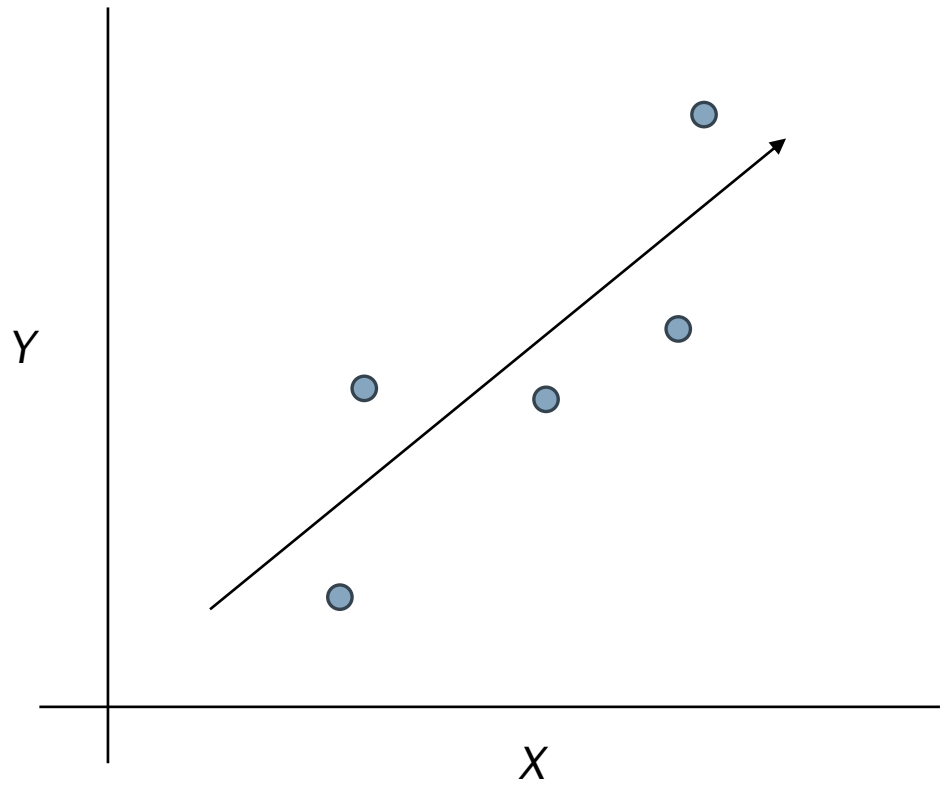
LOWESS fitting

StatQuest (https://statquest.org/)

https://youtu.be/Vf7oJ6z2LCc?si=WNOPz1hbbMhKt6bO

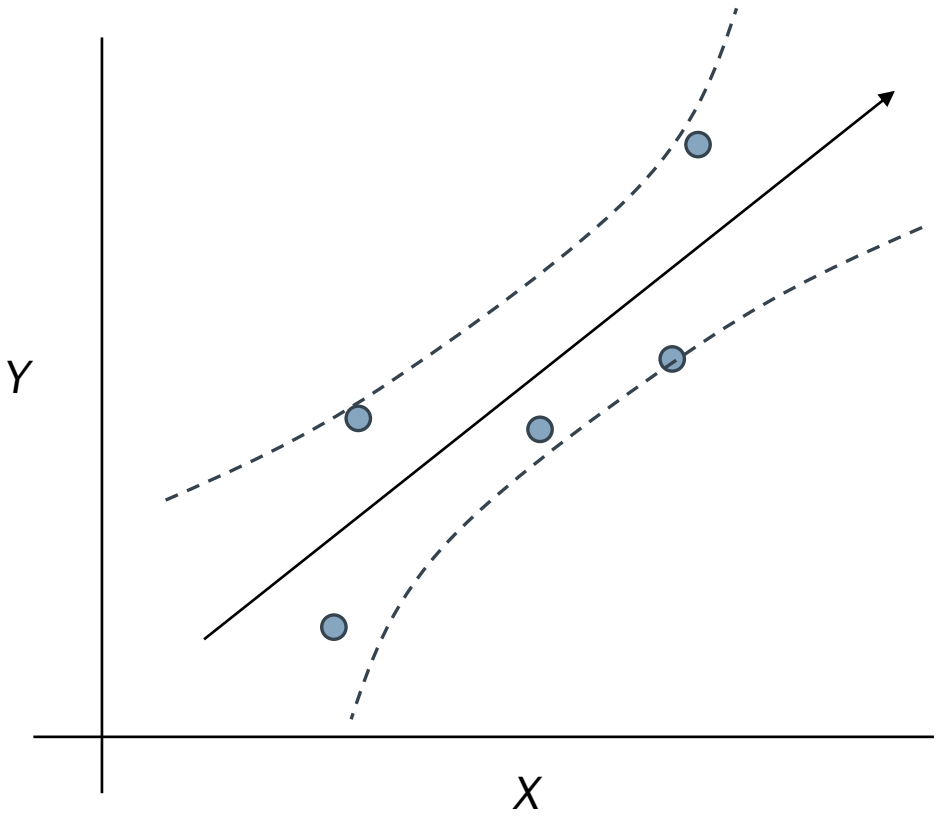
Parametric v.s. Nonparametric

- If you have a specific equation to represent the regression line/curve, you can call it "**parametric**".
  - Linear: $Y = aX + b$
  - Nonlinear: $Y = aX^2 + bX + c$
  - Nonlinear: Exponential, polynomial, …
- If not, you can call it "**nonparametric**".
  - Nonparametric does not assume any particular form (= a global equation that describes the data), so it can potentially generate linear and nonlinear regression.



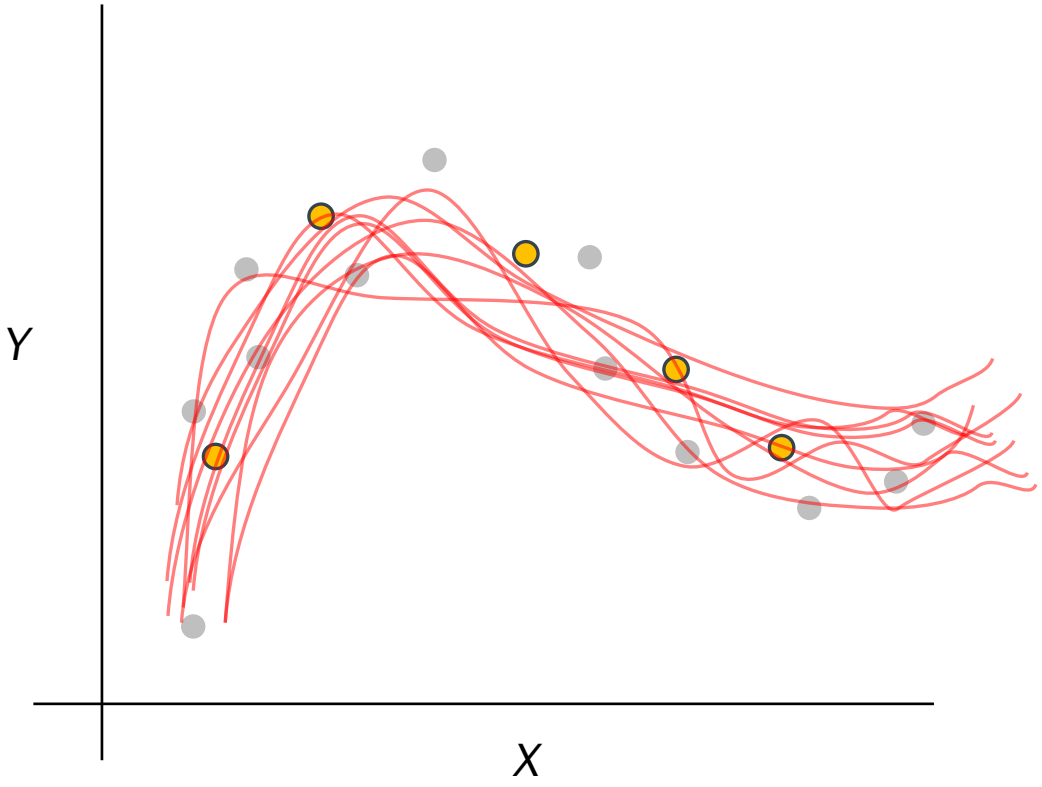
Confidence intervals (CIs)

- Confidence intervals indicate **how reliable the regression line/curve is.** Various methods are available.
- In the case of linear regression, variations (which are related to CIs) of a and b can be **mathematically estimated**.



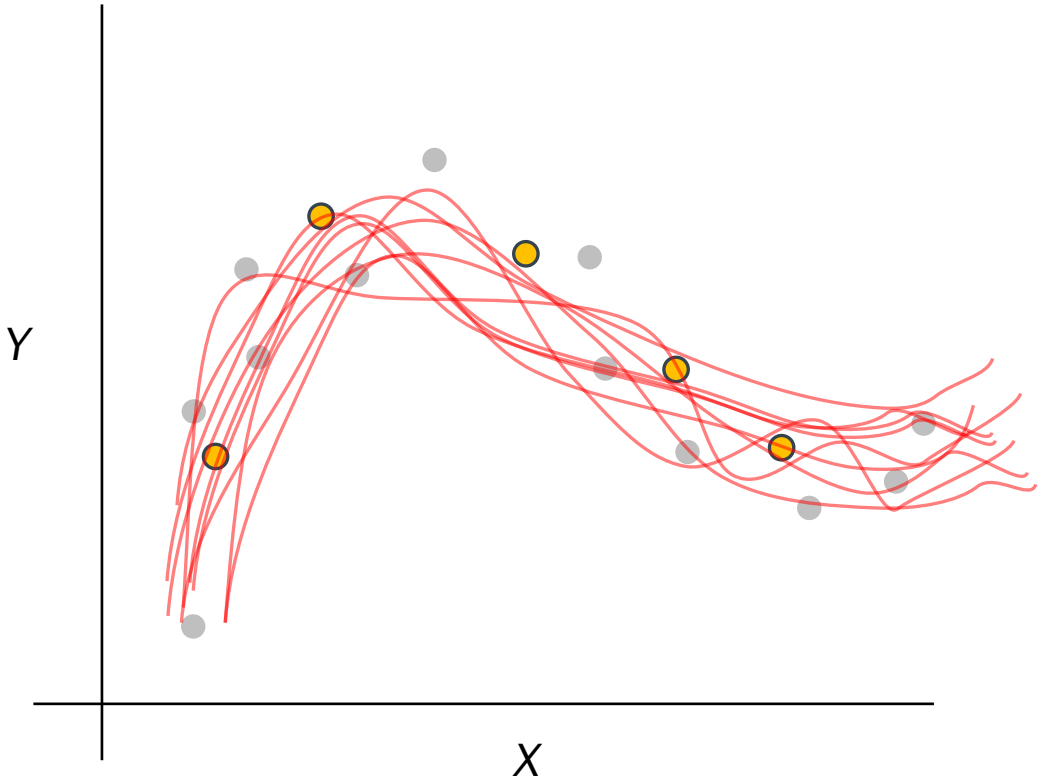
Confidence intervals (CIs)

- In the case of LOWESS, there is no explicit equation for the curve, but **the bootstrapping method can be used**.
- For example, we can randomly choose a subset of the original data (e.g., 50% of data), fit LOWESS curve, and repeat the process 100 (or more) times.
- You can extract 95 percentiles of the 100 LOESS lines to get 95% CIs.



Confidence intervals (CIs)

- NOTE: Bootstrapping methods include **random process(es)**. This means that you get a (slightly) different result each time you run the code.
- You need to set **a random seed** to reproduce the random process (= to get the same CIs).



Bootstrapping

StatQuest (https://statquest.org/)

https://youtu.be/Xz0x-8-cgaQ?si=wcMs1Inm8xVVHjBl

Confidence intervals (CIs)

StatQuest (https://statquest.org/)

https://youtu.be/TqOeMYtOc1w?si=qnAWHGsce7J4B3Zq

Other methods

- Many other methods are available to draw a regression line/curve.
- An important point (in this course) is **to understand which method you are using** and **explicitly state it in the figure legend**.
- **Linear regression** | The intercept should be 0 or not?; weighted linear regression; mixed linear model; standardized major axis (SMA)
- **Non-linear regression** | Nonparametric regression (no equation; e.g., LOWESS); polynomial regression (equation-based); Additive model
- **How to estimate parameters** | Least squared; Maximum likelihood; Bayesian estimation